

CE SECURITY

Security Operations Center

INCIDENT REPORT

IR-2026-001 | Linux Privilege Escalation Investigation

Severity: Medium | Status: Closed | Detection Source: Wazuh SIEM

Case Number	IR-2026-001
Classification	Internal Security Investigation
Analyst	Clinton Etienne
Organization	CE Security
Detection Platform	Wazuh SIEM
Endpoint	LINUX01
Report Version	1.0
Status	Closed

Executive Summary

A Wazuh alert identified suspicious privilege escalation activity on the Ubuntu endpoint LINUX01. The investigation revealed a sequence of administrative actions that included sudo execution, creation of a new account, escalation to the root user, and modification of account credentials.

Although this activity occurred in a controlled home lab environment, the event sequence closely mirrors attacker behavior commonly observed after successful credential compromise or unauthorized access. The investigation validated that Wazuh generated useful telemetry for Linux privilege escalation activity and provided evidence that could support triage, containment, and remediation in a production environment.

Incident Overview

Category	Value
Incident ID	IR-2026-001
Severity	Medium
Status	Closed
Detection Platform	Wazuh SIEM
Endpoint	LINUX01
Operating System	Ubuntu Linux
Analyst	Clinton Etienne
Framework	MITRE ATT&CK

MITRE ATT&CK Mapping

Tactic	Technique	Relevance
Privilege Escalation	T1548 - Abuse Elevation Control Mechanism	Use of sudo to obtain elevated permissions.
Persistence	T1136 - Create Account	Creation of an additional user account.
Credential Access	T1098 - Account Manipulation	Password modification and account-level changes.

Defense Evasion	T1078 - Valid Accounts	Activity performed using legitimate Linux account functionality.
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Investigation Timeline

Time	Event
09:17	User authenticated to Ubuntu endpoint LINUX01.
09:19	Sudo command executed.
09:20	Administrative account created.
09:21	Root shell obtained.
09:22	Password modified.
09:24	Wazuh generated privilege escalation alerts.
09:28	Investigation initiated.
09:40	Incident documented and closed.

Alert Evidence

Wazuh Alert

Rule: Privilege Escalation
User executed sudo
Severity: Medium
Endpoint: LINUX01
Status: Alert Generated

Account Creation

```
useradd analyst  
passwd analyst
```

Root Access

```
sudo su -  
whoami  
root
```

Password Change

```
passwd root
```

Analyst Findings

The investigation identified multiple privileged administrative actions occurring within a short time period. These actions included execution of sudo, new user creation, root shell acquisition, and password modification. Wazuh correlated the activity and generated alerts that provided useful evidence for triage.

In the home lab, the activity was authorized and performed for detection validation. In a production environment, this same sequence would require escalation because it may indicate compromised credentials, unauthorized administrative access, persistence creation, or preparation for further system modification.

- Sudo activity was observed on the Linux endpoint.
- A new account was created during the event window.

- Root-level access was obtained.
- Password modification activity occurred shortly after escalation.
- Wazuh produced alerts that supported incident investigation and documentation.

Risk Assessment

Risk Factor	Assessment
Likelihood	Medium
Impact	High
Overall Risk Rating	Medium
Primary Concern	Unauthorized administrative access and persistence.
Business Impact	Elevated access could allow system changes, data access, security control modification, or lateral movement.

Containment and Remediation

Completed Actions

- Verified that the activity originated from authorized home lab testing.
- Reviewed authentication and privilege escalation indicators.
- Validated that Wazuh captured the relevant Linux activity.
- Confirmed the endpoint remained available after testing.
- Documented the investigation using the CE Security incident report format.

Recommended Production Controls

- Implement least privilege for all Linux users.
- Restrict sudo permissions to approved administrative roles.
- Enable multi-factor authentication where supported.
- Alert on new account creation and password changes.
- Forward Linux authentication and audit logs to the SIEM.
- Review privileged group membership on a recurring schedule.

Lessons Learned

This investigation demonstrated the value of Linux endpoint monitoring through Wazuh. Privilege escalation activity can appear as legitimate system administration, so context is critical. Correlating sudo use, account creation, root access, and password changes helps determine whether the activity is authorized or potentially malicious.

Documenting the event from alert to closure strengthened incident response methodology, evidence handling, MITRE ATT&CK mapping, and SOC-style reporting. This report will serve as the master template for future CE Security investigations.

Skills Demonstrated

Skill Area	Demonstrated Capability
Linux security monitoring	Reviewed privilege escalation activity on an Ubuntu endpoint.
Wazuh SIEM	Validated alerts and supporting endpoint telemetry.
Incident response	Documented triage, findings, remediation, and lessons learned.
MITRE ATT&CK	Mapped observed activity to relevant tactics and techniques.
Technical documentation	Produced a professional incident report for portfolio review.